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# CS 621 DATA ANALYTICS

## Data Ingestion

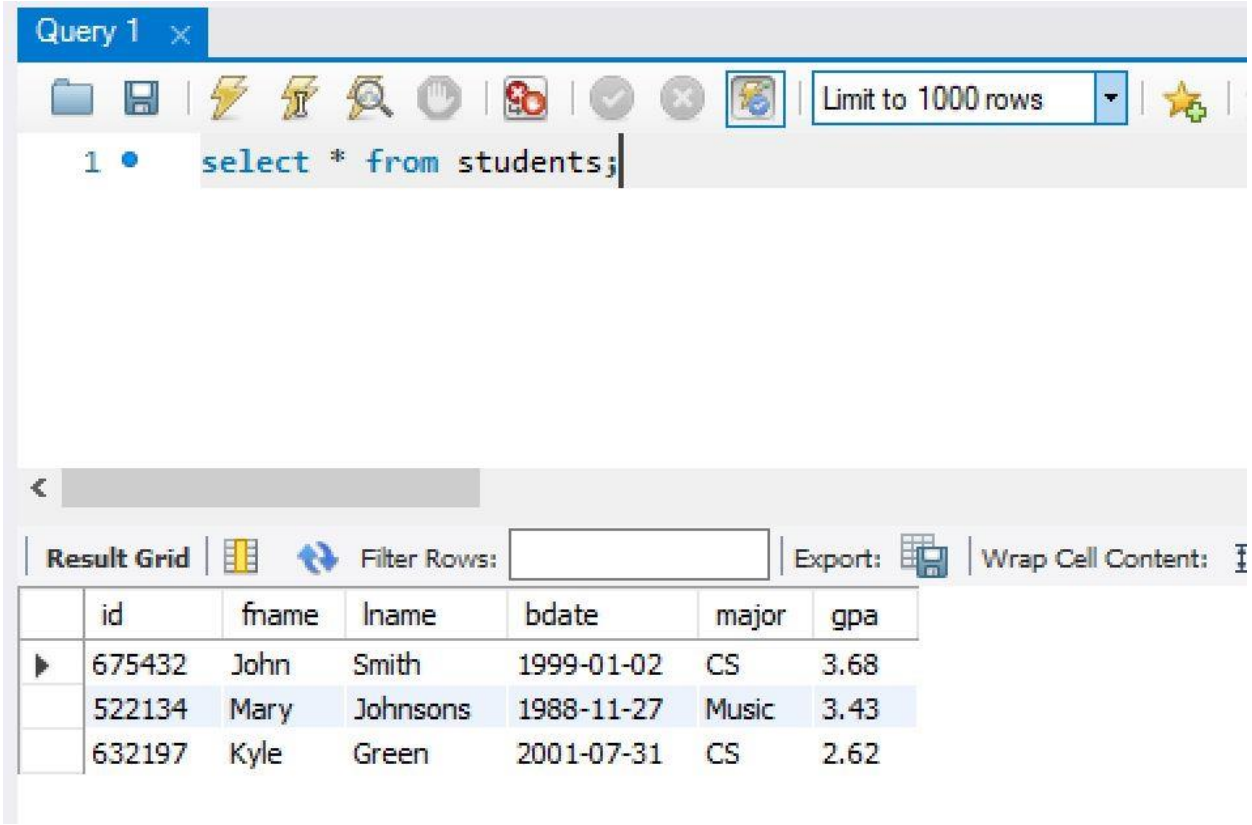
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# DATA STORAGES

- **Structured data** stored in databases, e.g., stored data in tables hosted in MySQL.
- **Semi-structured data** stored in JSON, XML, and CSV files.
- Data stored in regular text files (**unstructured data**).
- During the loading process, data would be stored into various data structures in the program.

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# A TABLE IN RELATIONAL DB




| id | fname | lname | bdate | major | gpa |
|---|---|---|---|---|---|
| 675432 | John | Smith | 1999-01-02 | CS | 3.68 |
| 522134 | Mary | Johnsons | 1988-11-27 | Music | 3.43 |
| 632197 | Kyle | Green | 2001-07-31 | CS | 2.62 |

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# DATA STORED IN RELATIONAL DB

- Data stored in MySQL database.
- >>> import MySQLdb
- >>> db = MySQLdb.connect(host=hostname, user=user_name, passwd=use_password, , db=database_name)
- >>> cur=db.cursor()
- >>> cur.execute("SELECT id, fname, lname FROM cs621")
- >>> for row in cur.fetchall():

  print(row)  # tuples

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# A JSON FILE EXAMPLE


```
[{"id":"675432", "fname":"John", "lname":"Smith", "bdate":"1999-01-02", "major":"CS"},
{"id":"522134", "fname":"Mary", "lname":"Johnsons", "bdate":"1988-11-27", "major":"Music"},
{"id":"632197", "fname":"Kyle", "lname":"Green", "bdate":"2001-07-31", "major":"CS"}]
```

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# DATA STORED IN JSON FORMAT

- Javascript Object Notation
- Name-value pair or key-value pair
- {“id”:”12345”}
- One json object defined by one pair {,,,,}
- Multiple json objects by [{}, {}, ]
- Examples
  - {“name”:”Sam”, “major”:“CS”, “age”:19}
  - [{“name”:”Sam”, “major”:“CS”, “age”:19}, {{“name”:”Bob”, “major”:“IT”, “age”:17}]

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# LOADING JSON FILES

- >>> import json
- >>> with open('students.json', 'r') as f:
    - students = json.load(f)
- The data structure of students can be:
    - a dictionary or
    - a list

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# LOAD JSON FILE INTO PANDAS

- We can also use Pandas for loading JSON files.
- import pandas as pd
- students = pd.read_json(“students.json”, orient=‘values’)

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# AN XML FILE EXAMPLE

```
<students>
        <student>
                <id>675432</id>
                <fname>John</fname>
                <lname>Smith</lname>
                <bdate>1999-01-02</bdate>
                <major> CS</major>
        </student>
        <student>
                <id>522134</id>
                <fname>Mary</fname>
                <lname>Johnsons</lname>
                <bdate>988-11-27</bdate>
                <major>Music</major>
        </student>
        <student>
                <id>632197</id>
                <fname>Kyle</fname>
                <lname>Green</lname>
                <bdate>2001-07-31</bdate>
                <major>CS</major>
        </student>
</students>
```

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# DATA STORED IN XML FORMAT

- Extensible Markup Language
- Have a root.
- Every element has a start tag, an end tag
- Need to be properly nested.
- See one example next.

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# LOADING XML DATA

- from xml.etree.ElementTree import parse
- doc = parse("the xml file")
- Use various methods/functions to retrieve the values of elements recursively. For example:

```
xml_data = []
for elem in root:
    temp = []
    for subelem in elem:
        temp.append(subelem.text)
    xml_data.append(temp)
```

- See code demonstration for more.

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# A CSV FILE EXAMPLE

```
Name, DOB, Major
Smith, 1999-01-02, CS
Johnsons, 1988-11-27, Music
Green, 2001-07-31, Biology
```

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# DATA IN CSV FORMAT

- Data fields are separated by ‘,’.
- Many business data stored in MS Excel can be viewed as csv files.
- A tabular spreadsheet looks like:

| Name | DOB | Major |
|---|---|---|
| Smith | 1999-01-02 | CS |
| Johnsons | 1988-11-27 | Music |
| Green | 2001-07-31 | Biology |

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# CSV FILE AS PLAIN TEXT

- Read each line in the CSV file.
- Separate each field/column using ‘,’.
- Convert data type if needed. For example:

  int(‘15’) will convert a string ’15’ into an integer 15.
- Store data in appropriate data structures.
- Similar approaches used on similar formats. For example, fields separated by single space, a tab \t, a vertical bar |, … etc.

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# LOAD CSV FILE INTO PANDAS DATAFRAME

- import pandas as pd
- df = pd.read_csv(“students.csv”)
- df is a dataframe.
- See code demonstration for more.

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# UNSTRUCTURED TEXT DATA

- Stored in text files locally or remotely.
- Need to retrieve/load the data first, process them, then store the processed data in different formats as discussed earlier, e.g., in a database, a json file, …, for future usage.
- Will discuss this more in text processing.



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